

Multimodal Bayesian Networks for Automatic Skin Disease Diagnosis

Yunqi He*

ECE Department

Northwestern University

Evanston, USA

yunqi.he@u.northwestern.edu

Jiahe Liu*

ECE Department

Northwestern University

Evanston, USA

jiahe.liu@u.northwestern.edu

Linglong Cai†

The Affiliated Hospital of
Guizhou Medical University

Guiyang, China

184112494c1@gmail.com

You Li

ECE Department

Northwestern University

Evanston, USA

you.li@u.northwestern.edu

Taimei Cui

The Affiliated Hospital of
Guizhou Medical University

Guiyang, China

cuitaimei@126.com

Hai Zhou

ECE Department

Northwestern University

Evanston, USA

haizhou@northwestern.edu

Abstract—Automatic diagnostic methods utilizing image analysis have seen a surge in development in recent years. However, achieving a comprehensive diagnosis necessitates considering clinical information. In the domain of skin disease, simply treating diagnosis as a multi-class image classification task poses inevitable bottlenecks. While existing methods incorporating Bayesian networks offer partial solutions, given the vast number of skin disease types, further integration with richer clinical information and diverse forms of deep neural networks would significantly escalate associated costs. This paper proposes an extensible diagnostic architecture. For newly added diseases and subsequently added clinical information, we integrate the output of deep neural networks into Bayesian networks more efficiently via semantic analysis. Through adjustments to node distribution types and the reduction of unnecessary connections, we expedite the development of a diagnostic system capable of addressing a wider range of skin diseases. Existing high-performing deep learning models, including multi-task and concept bottleneck, can be flexibly integrated into the system. Incorporating dermatologists' expertise and reusing trained neural networks additionally reduces system development costs. At the same time, our architecture ensures excellent diagnostic accuracy. Through experiments leveraging the PAD-UFES-20 and Skin-Con benchmarks, we have demonstrated that this architecture achieves a balanced diagnostic accuracy improvement of 19.3% over pure deep learning methods, with metrics including AUC surpassing those of existing published works. Our code can be accessed through the GitHub Repository: <https://github.com/KevinInfinigon/MultimodalBayesianNetworks>.

Index Terms—skin disease, multimodal, Bayesian network

I. INTRODUCTION

Skin cancer, primarily induced by exposure to harmful solar radiation, is the most prevalent type of cancer, leading to cell mutations, skin damage, even death. It encompasses three main types: Basal Cell Carcinoma (BCC), Squamous Cell Carcinoma (SCC), and the deadliest, Malignant Melanoma (MEL). Despite BCC and SCC being more common, they pose lower risks due to their limited metastatic potential,

whereas MEL, though least prevalent, accounts for a significant majority of skin cancer-related deaths due to its high metastatic propensity. Early detection of skin cancers is crucial for achieving optimal treatment outcomes, yet the clinical diagnostic approach heavily relies on inspection conducted by dermatologists using dermatoscopes, which is subjective and prone to inter- and intra-observer variability, demanding high specialty. Furthermore, the costliness and limited availability of dermatoscopes in resource-constrained regions hinder effective diagnosis.

To address these issues and improve diagnostic accuracy, particularly for early self-examination by patients in remote areas and supporting minimally trained dermatologists, computer-aided diagnosis (CAD) systems based on clinical image analysis have been developed [6]. Drawing inspiration from established dermatological criteria for diagnosing skin diseases, Yang et al. [26] devised effective visual medical representations to be extracted from clinical images. These extracted features serve as input for a random forest classifier, facilitating the classification of various skin lesions. Kalwa et al. [10] introduced a portable diagnostic system designed for smartphone use. This system utilizes smartphone cameras to capture clinical images, from which features corresponding to the ABCD rule [14] are extracted and inputted into a support vector machine to generate predictions regarding melanoma diagnosis.

With the rapid advancement of deep learning, significant breakthroughs continue to emerge. Shanthi et al. [22] utilized convolutional neural networks to extract learned features from clinical images, leveraging them for accurate classification predictions. Koh et al. [11] introduced Concept Bottleneck Models (CBMs), with the primary objective of augmenting the interpretability of deep learning models used in skin lesion classification tasks. Their approach involves training models to predict intermediate clinical concepts present in images rather than solely focusing on final diagnoses. Building upon this foundation, Yuksekgonul et al. [27] subsequently introduced

*Equal contribution.

†Linglong Cai is the corresponding author.

Post-hoc Concept Bottleneck Models (PCBMs). This novel methodology retains a high level of interpretability while simultaneously preserving the performance of deep learning models.

However, contemporary studies frequently focused on analyzing clinical images, overlooking the crucial clinical metadata associated with patients. In practice, dermatologists not only visually examine the patient's skin condition, but also inquire about the patient's demographic information in order to suggest a diagnosis. Recent trends in skin disease dataset curation, as highlighted in Wen et al.'s work [25], also indicate a growing inclusion of additional metadata alongside images, exemplified by datasets like PAD-UFES-20 [17]. This underscores the need for developing diagnostic systems based on multimodal fusion, which can effectively integrate both clinical images and patient metadata. In this context, Pacheco et al. [15] explored the impact of fusing metadata through concatenation. The embedded metadata vector is concatenated with the pooled-and-flattened extracted image features, ultimately achieving classification through a linear layer. However, this straightforward approach fails to adequately model the relationship between metadata and image features. To address this issue, Li et al. [12] proposed a multiplication-based fusion method, allowing direct interaction between image features and metadata. After transforming the embedded metadata vector to match the number of channels of the feature maps outputted by the final convolutional layer, it undergoes channel-wise multiplication with those feature maps. This process effectively reweights the feature maps with learned importance. Drawing inspiration from the gates in Long Short-Term Memory (LSTM) networks, Pacheco et al. [16] and Chen et al. [3] further proposed attention-based mechanisms to fully leverage metadata for guiding the selection of feature maps extracted from images.

Despite the success of the above-mentioned multimodal fusion techniques, there are still issues remain. Firstly, these methods essentially involve fitting a target function with multiple non-linear layers, which introduces additional overhead. If the conditional probability relationship between metadata and image information can be characterized, this overhead can then be avoided. Besides, the current use of metadata does not integrate the expert knowledge of dermatologists. However, it is clear that metadata can be more effectively leveraged in this manner. Furthermore, existing pure deep learning-based multimodal fusion methods lack sufficient interpretability, whereas interpretability of the model is crucial in clinical applications. Metadata itself carries semantics, and explaining its semantics in the context of the entire model would contribute to the development of a more explainable and reliable diagnostic system.

To this end, we propose employing multimodal Bayesian networks for the automatic diagnosis of skin diseases in this paper. Inspired from the construction of probabilistic graphical models (PGMs) with perception variables, hinge variables, and task variables in the field of Bayesian Deep Learning [24], we utilized Bayesian networks as the general framework to

integrate the outputs of deep learning models with other non-image metadata through multimodal fusion. During the construction of the multimodal Bayesian network, we utilized semantic analysis and the Bayesian formula to simplify the distribution types of the embedded deep learning nodes and reduce the number of connections. During training, the deep neural networks and Bayesian network were trained separately in a two-stage manner, making the system flexible. Due to the low cost of building Bayesian networks and their ability to leverage expert knowledge in constructing network structures, as well as the option to set prior probabilities to save on the required training data, we can rapidly train the Bayesian network and swiftly establish new diagnostic systems with agility. For newly added diseases, the concepts extracted using deep neural networks are consistent. Therefore, pre-trained deep learning models can be directly reused. Additionally, the parameters of the Bayesian network can be modified based on conditional probability relationships, without the need for additional training, enabling the assembly of a highly effective diagnostic system.

The remainder of the paper is structured as follows: Section II addresses the background of utilizing Bayesian networks for clinical diagnosis. Section III elaborates on our proposed architecture. Section IV provides a detailed explanation of the experimental process as well as an analysis of the experimental results. Section V summarizes, and outlines prospects for the entire research.

II. BACKGROUND

A. Bayesian Network-Driven Diagnostics

A Bayesian network [19], denoted as $G = \langle \mathbf{V}, \mathbf{E} \rangle$ ($|\mathbf{V}| = n$), is a directed acyclic graph depicting the dependencies among a set of random variables $\{X_i | i \in [1, n]\}$. The joint probability distribution of these random variables can be factorized as

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | Pa(X_i)), \quad (1)$$

where $Pa(X_i)$ denotes the parents of node X_i in the graph. The process of diagnosis naturally lends itself to modeling with Bayesian networks. Potential causes of disease, possible symptoms, and the presence of the disease itself can all be represented as different random variable nodes. Informally, we consider a factor to be the cause of a certain disease, and the former can be the parent node of the latter. Conveniently, the Bayesian formula

$$P(Y | X) = \frac{P(X | Y) \cdot P(Y)}{P(X)} \quad (2)$$

can be utilized to invert the prior and posterior probabilities. Hence, when constructing a network, the direction of edges can be adjusted based on the ease of obtaining relevant information.

We define a diagnostic Bayesian network $G_d = \langle \mathbf{V}_d, \mathbf{E}_d \rangle$, where $\mathbf{V}_d = \{C_i | i \in [1, m]\} \cup \{L\} \cup \{S_j | j \in [1, n]\}$ and $\mathbf{E}_d = \{\langle C_i, L \rangle | i \in [1, m]\} \cup \{\langle L, S_j \rangle | j \in [1, n]\}$. The

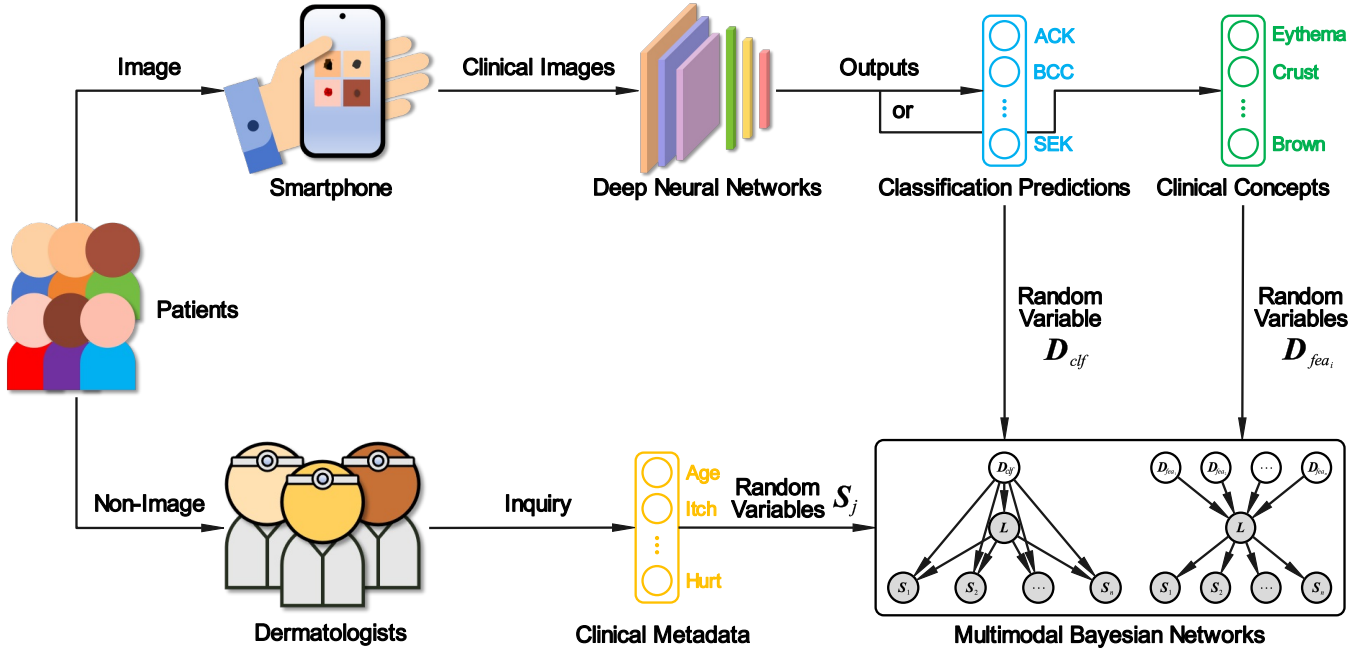


Fig. 1. Overview of the proposed multimodal Bayesian network architecture for skin disease diagnosis

node L represents the disease label, obeying a categorical distribution with k categories. C_i represents the i -th pathogenic factor and serves as the parent node of L . S_j stands for the j -th symptom and is a child node of L . When the p^{th} factor has value a and the q^{th} symptom has value b , we use

$$P(L | C_p = a, S_q = b) = \sum_{c \in \{C_i | i \neq p\}} \sum_{s \in \{S_j | j \neq q\}} \frac{P(C_p = a, c, L, S_q = b, s)}{P(C_p = a, S_q = b)} \quad (3)$$

to calculate the conditional distribution of the label L , that is, to give a diagnosis. This formula can be further broken down through the chain rule and quickly solved using the variable elimination algorithm. When all nodes except L are given values, we can get a very accurate diagnostic result.

It's worth noting that the absence of certain node information doesn't hinder the computation. This mirrors real-world diagnosis scenarios where complete patient information is often unavailable. We can prioritize completing missing data based on their impact on the marginal probability distribution. In medical contexts, this often guides the recommendation of necessary patient examinations.

B. Related Work

Early attempts to utilize Bayesian networks for organizing the output generated by neural networks have been documented. For instance, [20] attempted to amalgamate the output of four simple visual algorithms to detect the presence of a speaker. In a similar vein, [21] employed deep learning techniques to extract features from brain MRI images and utilized Bayesian networks for subprofessional-level brain lesion diagnosis. Notably, the deep learning models utilized in this study were specialized feature recognition networks,

and the approach did not delve into the utilization of general deep learning models for inference.

In another study, [9] investigated the integration of deep neural networks and Bayesian networks for skin disease diagnosis. The novelty of this approach lies in modeling the output of a general neural network as a set of Gaussian random variables, akin to human expert inspection. Characterizing the mean and variance separately for each dimension of the neural network output resulted in a doubling of the number of parameters requiring training, along with additional sampling overhead. This approach, while aligned with dermatologists' considerations for skin disease diagnosis, presents difficulties when applied to a potential large k -category diagnosis system.

Assuming m Bernoulli nodes in the first layer and n Bernoulli nodes in the third layer, the entire network requires establishing $(k + 1)(m + n)$ connections and a total of $k \cdot 2^{m+n} + 2^m + kn$ distribution functions, all of which require parameter training. Consequently, this suggests that the architecture may face limitations when dealing with an increased number of non-image information nodes and a broader spectrum of diseases.

III. METHOD

In response to the challenges posed by the existing Bayesian network-driven diagnostic architecture, we first classified the deep learning models integrated into the system based on their output types. This classification laid the foundation for restructuring the approach to incorporating deep learning nodes into the Bayesian network, as detailed in Section III-A. Building upon this improved incorporation method, we further devised more efficient connection strategies in Section III-B, aiming to reduce the overall count of connections and enhance the system's performance. These design improvements, in turn,

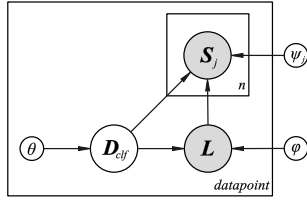


Fig. 2. Architecture of the classification-based Multimodal Bayesian Network (D_{clf} system). Note: D_{clf} represents the deep learning node that outputs classification predictions, L is the disease label node, and S_j ($j = 1, 2, \dots, n$) are the metadata nodes.

enabled us to introduce an agile system extension mechanism in Section III-C, allowing for the seamless integration of new information nodes and the expansion of the system's diagnostic range.

Fig. 1 provides an overview of our proposed multimodal Bayesian network architecture, illustrating the integration of image feature extraction, metadata processing, and Bayesian network components for skin disease diagnosis.

A. Deep Learning as a Node

The output of deep learning models used in medical diagnosis typically manifests in two primary forms: classification predictions based on input images [22], and evaluation of specific image features [11].

For the former, the output layer of the deep neural network commonly employs a fully connected layer activated by the softmax function. The linear layer computes the unnormalized log probabilities $z = W^T h + b$, where $z_i = \log \hat{P}(y = i | x)$. Subsequently, the softmax function exponentiates and normalizes z to derive the desired probabilities $\hat{y}_i = P(y = i | x)$. The softmax function is defined as $\text{softmax}(z)_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$. The output vector $\text{softmax}(z)$ can just be used as the parameter of a categorical distribution. Its probability mass function is $f(x; z) = \prod_{i=1}^k z_i^{[x=i]}$, where $[x = i]$ evaluates to 1 if $x = i$, 0 otherwise. We connect a random variable obeying this categorical distribution as a node into the Bayesian network.

For the latter, the output layer of the neural network corresponding to each image feature is $\hat{y} = \sigma(w^T h + b)$, where σ denotes the sigmoid function, which maps the linear layer's output to a probability $\hat{y} \in [0, 1]$, indicating the likelihood of the presence of a high-level clinical concept. Similarly, we incorporate random variables into the Bayesian network, each following a Bernoulli distribution $f(k; \hat{y}) = \hat{y}^k (1 - \hat{y})^{1-k}$, for $k \in \{0, 1\}$.

B. Efficient Connection

As mentioned earlier, leveraging the flexibility provided by formula 2, the connections within the Bayesian network are inherently bidirectional. Guided by experts' recommendations and the feature screening outcomes generated by the random forest algorithm, we have retained the essential connections. Additionally, we adjusted the direction of these connections, aiming to minimize their overall count. For the two types of deep learning models—classification prediction and image

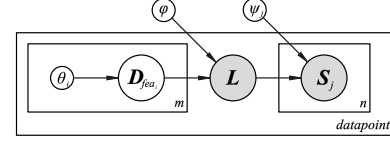


Fig. 3. Architecture of the feature-based Concept Bayesian Network (D_{fea} system). Note: The figure shows a single D_{fea} node for simplicity, but in practice, multiple D_{fea} nodes ($i = 1, 2, \dots, m$) connect to the label node L .

feature evaluation—diagnostic systems are connected slightly differently. Commonly, for a k -category disease diagnosis problem, both types involve a disease label node L that follows a categorical distribution with k categories, as well as a series of metadata nodes $S_j, j \in [1, n]$ with different distributions. Naturally, $\forall j \in [1, n], \exists \langle L, S_j \rangle$.

For the former, classification prediction gives a comprehensive judgment. The deep learning node D_{clf} that obeys a k -category categorical distribution needs to be connected to L and all $S_j, j \in [1, n]$. We arrange D_{clf} as the parent node of L and all $S_j, j \in [1, n]$, and L as the parent node of all $S_j, j \in [1, n]$, so that the number of connections in the network is $2n + 1$. The data contained in the L node is $P(L | D_{\text{clf}})$, which is k different distribution functions. The data contained in S_j node is $P(S_j | L, D_{\text{clf}})$, which is k^2 different distribution functions. The total number of distribution functions that require parameter training in this Bayesian network is $nk^2 + k$.

For the latter, image feature evaluation generally provides information about specific high-level clinical concepts, and their direct correlation with metadata is weak. Therefore, we consider a set of deep learning nodes $D_{\text{fea}}, i \in [1, m]$ that each obeys a Bernoulli distribution. We arrange each $D_{\text{fea}}, i \in [1, m]$ as the parent node of L , and L as the parent node of all $S_j, j \in [1, n]$, so that the number of connections in the network is $m + n$. The data contained in the L node is $P(L | D_{\text{fea}_1}, \dots, D_{\text{fea}_m})$, which is 2^m different distribution functions. The data contained in S_j node is $P(S_j | L)$, which is k different distribution functions. The total number of distribution functions that require parameter training in this Bayesian network is $2^m + k$.

The architectures of the D_{clf} and D_{fea} systems are illustrated in Fig. 2 and Fig. 3, respectively. Regarding the distribution parameters of D_{clf} and $D_{\text{fea}}, i \in [1, m]$, they are directly derived from the trained neural network. These parameters remain fixed during the training stage of the Bayesian network, hence, they are not counted among the distributions that require training. Our approach demonstrates superiority over the existing method mentioned in Section II in terms of network connectivity and the scale of training parameters.

C. Architecture Extensibility

1) *Node Addition*: As the data set expands and enriches, we may get new categories of metadata. S_{new} denotes the node corresponding to the newly added category of metadata.

For the D_{clf} system, the Bayesian network needs to add 2 edges: $\langle D_{\text{clf}}, S_{\text{new}} \rangle$ and $\langle L, S_{\text{new}} \rangle$. $P(S_{\text{new}} | L, D_{\text{clf}})$ attached to S_{new} node is k^2 different distribution functions. For the D_{fea} system, the Bayesian network needs to add 1 edge: $\langle L, S_{\text{new}} \rangle$. The data contained in S_{new} node is $P(S_{\text{new}} | L)$, which is k different distribution functions. Due to independence, only the parameters of the newly added distribution function need to be trained.

In particular, for D_{fea} system, when new image feature node D'_{fea} is added, edge $\langle D'_{\text{fea}}, L \rangle$ needs to be added. At this time, the data contained in the L node needs to be replaced with new 2^{m+1} distribution functions. However, there is no need to retrain the distributions of other nodes.

2) *Diagnostic Range Extension*: A commonly encountered and challenging scenario arises when a trained k -category diagnosis system needs to acquire the capability to diagnose an additional l diseases. In contrast to a complete redesign, our architecture can seamlessly extend the original system to accommodate a $k + l$ -category system.

For the D_{fea} system, the high-level clinical concepts remain consistent with existing ones even for newly added diseases, enabling the reuse of the image feature networks without the need for retraining. Therefore, we can retain all $D_{\text{fea}_i}, i \in [1, m]$ from the original system and maintain the parameters of their distribution functions unchanged. We can train other parts of the Bayesian network only for the newly added l diseases, and then merge with the k -category system at a constant cost to form the target system. The parameter vector for the label node L_{ori} in the original system is denoted as φ_{ori} , while that for the l -category system is denoted as φ_l . The parameter vector for the label node L_{new} in the target system is denoted as φ_{new} , which is obtained through weighted concatenation:

$$\varphi_{\text{new}} = \left\{ \frac{k}{k+l} \cdot \varphi_{\text{ori}}, \frac{l}{k+l} \cdot \varphi_l \right\} \quad (4)$$

During merging, the distribution functions of the same meta-data nodes from both the original system and the l -category system are straightforwardly combined to form the corresponding nodes of the target system. This is because the metadata nodes contain independent distribution functions.

IV. EXPERIMENT

All our experiments were carried out using PyTorch [18] deep learning framework and Pyro [2] deep probabilistic programming framework on a single NVIDIA RTX A6000 GPU (48 GB).

A. Datasets

We have leveraged two skin disease benchmark datasets to train and evaluate our models: the PAD-UFES-20 dataset and the SkinCon [4] dataset.

PAD-UFES-20 This metadata-rich dataset consists of 2298 clinical images over 6 different skin diseases: 3 benign diagnoses - Actinic Keratosis (ACK), Melanocytic Nevus (NEV), and Seborrheic Keratosis (SEK); 3 malignant diagnoses: Basal Cell Carcinoma (BCC), Malignant Melanoma (MEL), and

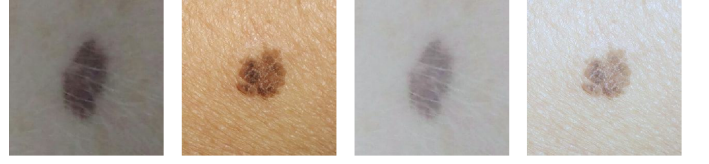


Fig. 4. Comparison between resized skin lesion images (left half) and images pre-processed with Shades of Gray color constancy algorithm (right half).

Squamous Cell Carcinoma (SCC). Each skin lesion image is provided with up to 21 clinical features: age, region, diameter, itch, grew, hurt, *etc.*. The PAD-UFES-20 dataset is available for download at: <https://data.mendeley.com/datasets/zr7vgbcyr2/1>.

SkinCon This densely annotated dataset mainly consists of 3230 clinical images taken as a subset of the Fitzpatrick-17K dataset [8]. Each skin lesion image is labeled with up to 48 clinical concepts by dermatologists: erythema, crust, telangiectasia, nodule, *etc.*. The SkinCon dataset is available for download at: <https://skincon-dataset.github.io/>.

B. Implementation Details

1) *Image Pre-Processing*: To be compatible with the input size of the chosen deep neural network backbone - Efficient-NetB3 [23], skin lesion images from both PAD-UFES-20 and SkinCon datasets were resized to 300×300 pixels.

Color Constancy Taking into account that skin lesion images were captured under a wide variety of lighting conditions, and the skin tone also differs greatly across image samples, we applied the Shades of Gray [7] color constancy algorithm to normalize the colors, which has been proven to improve diagnostic accuracy by Barata et al. [1]. More specifically, this process constitutes a gamma correction step with $\gamma = 2.2$, followed by a color constancy step with $p = 6$. The effect of this process is illustrated in Fig. 4.

Data Augmentation Regarding the diagnoses and concepts as class labels, the distribution of classes is highly skewed for both PAD-UFES-20 and SkinCon datasets. In order to prevent the over-fitting of deep neural networks, we employed heavy data augmentation techniques. The random transformations used for data augmentation during training are: transpose, horizontal and vertical flipping, affine transforms, cropping, change in HSV values, change in brightness and contrast, blurs and distortions, and contrast limited adaptive histogram equalization (CLAHE).

2) *Metadata Pre-Processing*: For the PAD-UFES-20 dataset, we have selected 9 out of the available 21 clinical features as Bayesian network nodes/variables. This decision was guided by the outcomes of exploratory data analysis and enriched by insights obtained from collaborating dermatologists.

Binary Features The 6 original binary features are: itch, grew, hurt, changed, bleed, elevation. Among these features, about 6% of data entries are missing. Thus, k-Nearest Neighbor imputation with $k = 1$ was performed to compute the missing values for each feature.

Categorical Feature The 1 original categorical feature is: region. This feature has no missing data entries, thus feature transformation was performed by simply sorting the string values in alphabetical order, followed by replacement with consecutive integers, resulting in 14 distinct categories.

Numerical Features The 2 original numerical features are: age, diameter. The ‘age’ feature has complete data entries and was discretized into a categorical feature by employing a binning interval of 10, resulting in 10 distinct categories. In contrast, the ‘diameter’ feature was derived by applying a maximum operation to the original horizontal and vertical diameters. Due to its 35% missing data entries, k-Nearest Neighbor imputation with $k = 5$ was performed. Subsequently, discretization was carried out by setting the binning interval to $5mm$ for diameters up to $35mm$, while the remaining diameters were all binned into one single category, resulting in 8 distinct categories in total.

3) *Model Training and Evaluation:* For experiments conducted on the PAD-UFES-20 dataset, we employed leak-free stratified 5-fold cross-validation to ensure robustness across different train-val splits. As for experiments leveraging SkinCon, a stratified train-val split of 4 : 1 was applied to this dataset. Eventually, all models were evaluated on the PAD-UFES-20 hold-out test set. The mean and standard deviation of multiclass accuracy (ACC), balanced multiclass accuracy (BACC), and macro-averaged AUROC (AUC) over all folds were reported for each model. Since the dataset is highly imbalanced, we consider BACC as the main ranking metric.

Image-Only Deep Neural Network Our first baseline architecture is an EfficientNetB3 convolutional neural network with skin lesion images as the only source of input. The model was initialized with ImageNet [5] pre-trained weights, and further fine-tuned on the PAD-UFES-20 dataset. Apart from applying data augmentation, a focal loss [13] weighted by the inverse class frequencies was adopted to further ameliorate the data imbalance issue. The Adam optimizer with a batch size of 64 was used, along with the learning rate starting from 1×10^{-5} , linearly warm up to 1×10^{-4} at 10^{th} epoch, then cosine-anneal to 1×10^{-5} at the final 100^{th} epoch. Early-stopping regularization based on validation BACC was adopted for model selection.

Metadata-Only Bayesian Network Our second baseline architecture is a Bayesian network constructed with variables only representing diagnostic labels and patient clinical metadata. All variables are observed, categorically distributed, and given non-informative priors, which assigns equal probabilities to all categories. Maximum likelihood estimation was performed for parameter learning by optimizing the evidence lower bound (ELBO) loss with the Adam optimizer leveraging the PAD-UFES-20 dataset. A batch size of 64 was used, along with a learning rate of 2.5×10^{-3} over 100 epochs. Early-stopping regularization based on validation BACC was adopted for model selection.

Multimodal Bayesian Network Our proposed architecture based on the D_{cif} system. Skin lesion images are first processed through a trained deep neural network backbone, and

the softmax-ed outputs are then embedded into a Bayesian network as the parameters of a latent categorical variable, connected with other observed categorical variables representing diagnostic labels and patient clinical metadata. More specifically, the above-mentioned trained Image-Only DNN was selected as the backbone and had all parameters frozen. The rest of the variables were initialized and trained in the same manner on the PAD-UFES-20 dataset as the above-mentioned Metadata-Only BN.

Concept Bayesian Network Our proposed architecture based on the D_{fea} system. Instead of representing the classification predictions of the deep neural network with a single latent categorical variable, multiple latent Bernoulli variables with each representing a clinical concept were used. Taking the intersection of the top-ranked concepts obtained from a random forest classifier and that given by collaborating dermatologists, we have chosen 5 out of the available 48 clinical concepts: erythema, crust, telangiectasia, brown, black. Correspondingly, multiple EfficientNetB3 deep neural networks were trained on the SkinCon dataset with each responsible for extracting a single clinical concept from the skin lesion images. These trained backbones were subsequently frozen and utilized for processing skin lesion images from the PAD-UFES-20 dataset to embed them into the desired latent variables, which were then integrated with additional observed variables whose parameters were learned with diagnostic labels and patient clinical metadata. The training details remain consistent with those outlined previously.

In order to demonstrate the extensibility of this proposed architecture, a follow-up experiment was also conducted. The 6 diagnostic classes of PAD-UFES-20 were split into two groups, with the first group containing 4 classes: ACK, BCC, MEL, NEV; and the second group containing 2 classes: SCC, SEK. The deep neural network backbones were repurposed, with their trained parameters retained, while two separate Bayesian networks were subsequently trained for the classification of 4 and 2 classes, respectively. Following their individual training phases, the parameters of these Bayesian networks were fused without undergoing additional training iterations to classify the entire 6 classes. More specifically, the fusion of the label variable L parameters involved weighting by the number of classes followed by concatenation, whereas the fusion of metadata variables $S_j, j \in [1, n]$ parameters entailed concatenation alone.

C. Experimental Results

1) *Ablation Study:* To evaluate the effectiveness of our approach, we compared our proposed architectures with two baseline architectures. The first baseline architecture is the Image-Only DNN, and the second baseline architecture is the Metadata-Only BN. As depicted in Table I, these models which were trained solely on single-modality data produce results that are deemed acceptable yet fall short of expectations. Importantly, all metrics of the Metadata-Only BN surpass that of the Image-Only DNN by a large margin, underscoring the diagnostic efficacy inherent in clinical in-

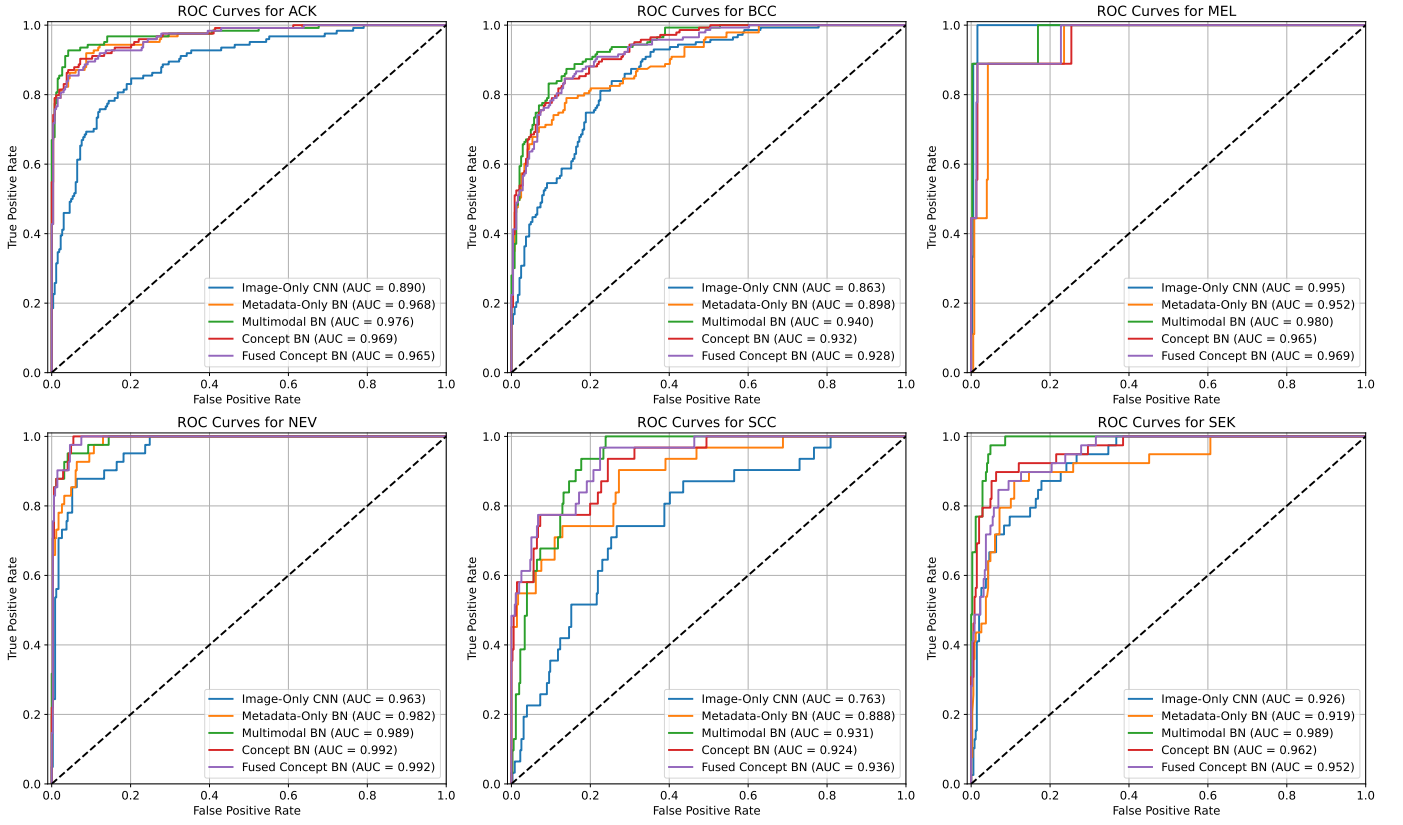


Fig. 5. ROC curves for six skin diseases across five model architectures. The Multimodal Bayesian Network (green) performs excellently in the diagnosis task across all six disease classes.

TABLE I
PERFORMANCE COMPARISON OF PROPOSED ARCHITECTURES WITH
BASELINE METHODS (MEAN $\pm$ STD).

Approach	ACC	BACC	AUC
Image-Only CNN	0.642 $\pm$ 0.024	0.618 $\pm$ 0.013	0.902 $\pm$ 0.003
Metadata-Only BN	0.744 $\pm$ 0.008	0.661 $\pm$ 0.017	0.930 $\pm$ 0.005
Concept BN	0.784 $\pm$ 0.006	0.771 $\pm$ 0.008	0.953 $\pm$ 0.003
Multimodal BN	0.824 $\pm$ 0.011	0.811 $\pm$ 0.020	0.965 $\pm$ 0.005

TABLE II
EXTENSIBILITY ANALYSIS: COMPARISON OF DIRECTLY TRAINED AND
FUSED CONCEPT BAYESIAN NETWORKS (MEAN $\pm$ STD).

Approach	ACC	BACC	AUC
Concept BN	0.784 $\pm$ 0.006	0.771 $\pm$ 0.008	0.953 $\pm$ 0.003
Fused Concept BN	0.747 $\pm$ 0.006	0.770 $\pm$ 0.008	0.951 $\pm$ 0.004

formation. With the incorporation of multimodal fusion techniques, our proposed Multimodal BN and Concept BN exhibit a notable enhancement in performance when compared to the baseline architectures, achieving a BACC improvement of 19.3% and 15.3% over Image-Only DNN, 15.0% and 11.0% over Metadata-Only BN, respectively. This demonstrates that our proposed architectures ensure excellent diagnostic performance while retaining the high explainability inherent in Bayesian networks. Fig. 5 presents the ROC curves for all six diseases across the five model architectures. The Multimodal BN consistently achieves high AUC scores, demonstrating its superior diagnostic capability.

2) *Extensibility Analysis:* To assess the extensibility of our approach, we conducted a comparative analysis between the Concept BN trained to directly classify the 6 diagnostic classes within the PAD-UFES-20 dataset and a Fused Concept BN, which was created by fusing the parameters of two Bayesian networks, each trained to classify a non-overlapping subset of the diagnostic classes. As depicted in Table II, the Fused Concept BN demonstrates the ability to attain comparable BACC and AUC metrics with the original Concept BN, albeit at the expense of compromised ACC. This illustrates that the clinical concepts and features we have selected exhibit a high degree of representativeness, thereby rendering the Bayesian networks built with these variables extensible. While a single multi-task model or concept bottleneck model which offers a uniform solution can be seamlessly integrated into this architecture, we have opted to train individual deep learning models for each concept. This ensures that these models can be readily repurposed when new concepts emerge.

3) *Comparison with Existing Approaches:* As shown in Table III, it is evident that our proposed architectures consistently achieve high rankings compared to existing approaches. Particularly noteworthy is the performance of Multimodal BN, which emerges as the frontrunner. Additionally, Concept BN demonstrates commendable and comparable results, underscoring its efficacy and extensibility. It's important to note that due to the absence of an official test set split for the

TABLE III
COMPARISON OF PROPOSED METHODS WITH STATE-OF-THE-ART
APPROACHES FOR SKIN DISEASE DIAGNOSIS.

Approach	Mean ACC	Mean BACC	Mean AUC
Pacheco et al. [15]	0.712 ~ 0.765	0.720 ~ 0.758	0.927 ~ 0.945
Li et al. [12]	0.700 ~ 0.749	0.717 ~ 0.754	0.922 ~ 0.937
Pacheco et al. [16]	0.723 ~ 0.748	0.736 ~ 0.770	0.931 ~ 0.944
Chen et al. [3]	0.804	0.800	0.950
Concept BN	0.784	0.771	0.953
Multimodal BN	0.824	0.811	0.965

PAD-UFES-20 dataset, the reported results shall only serve as a reference point for comparison.

V. CONCLUSION

This paper proposes a novel multimodal Bayesian network architecture for automatic skin disease diagnosis, effectively combining deep learning models with Bayesian networks to comprehensively process both image and non-image information. By classifying deep learning output types and restructuring the incorporation method, we reduce the number of connections and distribution functions, enabling an extensible diagnostic system. Experiments on PAD-UFES-20 and SkinCon demonstrate superior performance, with the Multimodal Bayesian Network improving balanced accuracy by 19.3% over pure deep learning methods. The Concept Bayesian Network also exhibits commendable results and extensibility. Future work will focus on further enhancing the architecture's adaptability, exploring advanced deep learning models, and investigating the possibility of integrating large vision-language models to assist in diagnosis.

REFERENCES

- [1] Catarina Barata, M Emre Celebi, and Jorge S Marques. Improving dermoscopy image classification using color constancy. *IEEE journal of biomedical and health informatics*, 19(3):1146–1152, 2014.
- [2] Eli Bingham, Jonathan P Chen, Martin Jankowiak, Fritz Obermeyer, Neeraj Pradhan, Theofanis Karaletsos, Rohit Singh, Paul Szerlip, Paul Horsfall, and Noah D Goodman. Pyro: Deep universal probabilistic programming. *The Journal of Machine Learning Research*, 20(1):973–978, 2019.
- [3] Qian Chen, Min Li, Chen Chen, Panyun Zhou, Xiaoyi Lv, and Cheng Chen. Mdfnet: application of multimodal fusion method based on skin image and clinical data to skin cancer classification. *Journal of Cancer Research and Clinical Oncology*, 149(7):3287–3299, 2023.
- [4] Roxana Daneshjou, Mert Yuksekgonul, Zhuo Ran Cai, Roberto Novoa, and James Y Zou. Skincon: A skin disease dataset densely annotated by domain experts for fine-grained debugging and analysis. *Advances in Neural Information Processing Systems*, 35:18157–18167, 2022.
- [5] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009.
- [6] Luigi Di Biasi, Alessia Auriemma Citarella, Michele Risi, and Genoveffa Tortora. A cloud approach for melanoma detection based on deep learning networks. *IEEE Journal of Biomedical and Health Informatics*, 26(3):962–972, 2021.
- [7] Graham D Finlayson and Elisabetta Trezzi. Shades of gray and colour constancy. In *Color and Imaging Conference*, volume 12, pages 37–41. Society of Imaging Science and Technology, 2004.
- [8] Matthew Groh, Caleb Harris, Luis Soenksen, Felix Lau, Rachel Han, Aerin Kim, Arash Koochek, and Omar Badri. Evaluating deep neural networks trained on clinical images in dermatology with the fitzpatrick 17k dataset. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1820–1828, 2021.
- [9] Yunqi He, Linglong Cai, Taimei Cui, You Li, and Hai Zhou. A combination of dnn and bn for automatic skin disease diagnosis. In *2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI)*, pages 1–5. IEEE, 2023.
- [10] Upender Kalwa, Christopher Legner, Taejoon Kong, and Santosh Pandey. Skin cancer diagnostics with an all-inclusive smartphone application. *Symmetry*, 11(6):790, 2019.
- [11] Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. Concept bottleneck models. In *International conference on machine learning*, pages 5338–5348. PMLR, 2020.
- [12] Weipeng Li, Jiaxin Zhuang, Ruixuan Wang, Jianguo Zhang, and Wei-Shi Zheng. Fusing metadata and dermoscopy images for skin disease diagnosis. In *2020 IEEE 17th international symposium on biomedical imaging (ISBI)*, pages 1996–2000. IEEE, 2020.
- [13] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision*, pages 2980–2988, 2017.
- [14] Franz Nachbar, Wilhelm Stolz, Tanja Merkle, Armand B Coggnetta, Thomas Vogt, Michael Landthaler, Peter Bilek, Otto Braun-Falco, and Gerd Plewig. The abcd rule of dermatoscopy: high prospective value in the diagnosis of doubtful melanocytic skin lesions. *Journal of the American Academy of Dermatology*, 30(4):551–559, 1994.
- [15] Andre GC Pacheco and Renato A Krohling. The impact of patient clinical information on automated skin cancer detection. *Computers in biology and medicine*, 116:103545, 2020.
- [16] Andre GC Pacheco and Renato A Krohling. An attention-based mechanism to combine images and metadata in deep learning models applied to skin cancer classification. *IEEE journal of biomedical and health informatics*, 25(9):3554–3563, 2021.
- [17] Andre GC Pacheco, Gustavo R Lima, Amanda S Salomao, Breno Krohling, Igor P Biral, Gabriel G de Angelo, Fábio CR Alves Jr, José GM Esgario, Alana C Simora, Pedro BC Castro, et al. Pad-ufes-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones. *Data in brief*, 32:106221, 2020.
- [18] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style, high-performance deep learning library. *Advances in neural information processing systems*, 32, 2019.
- [19] Judea Pearl. *Probabilistic reasoning in intelligent systems: networks of plausible inference*. Morgan kaufmann, 1988.
- [20] James M Rehg, Kevin P Murphy, and Paul W Fieguth. Vision-based speaker detection using bayesian networks. In *Proceedings. 1999 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (Cat. No PR00149)*, volume 2, pages 110–116. IEEE, 1999.
- [21] Jeffrey D Rudie, Andreas M Rauschecker, Long Xie, Jiancong Wang, Michael Tran Duong, Emmanuel J Botzakis, Asha Kovalovich, John M Egan, Tessa Cook, R Nick Bryan, et al. Subspecialty-level deep gray matter differential diagnoses with deep learning and bayesian networks on clinical brain mri: a pilot study. *Radiology: Artificial Intelligence*, 2(5):e190146, 2020.
- [22] T Shanthi, RS Sabeenian, and R Anand. Automatic diagnosis of skin diseases using convolution neural network. *Microprocessors and Microsystems*, 76:103074, 2020.
- [23] Mingxing Tan and Quoc Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *International conference on machine learning*, pages 6105–6114. PMLR, 2019.
- [24] Hao Wang and Dit-Yan Yeung. A survey on bayesian deep learning. *ACM computing surveys (csur)*, 53(5):1–37, 2020.
- [25] David Wen, Saad M Khan, Antonio Ji Xu, Hussein Ibrahim, Luke Smith, Jose Caballero, Luis Zepeda, Carlos de Blas Perez, Alastair K Denniston, Xiaoxuan Liu, et al. Characteristics of publicly available skin cancer image datasets: a systematic review. *The Lancet Digital Health*, 4(1):e64–e74, 2022.
- [26] Jufeng Yang, Xiaoxiao Sun, Jie Liang, and Paul L Rosin. Clinical skin lesion diagnosis using representations inspired by dermatologist criteria. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 1258–1266, 2018.
- [27] Mert Yuksekgonul, Maggie Wang, and James Zou. Post-hoc concept bottleneck models. *arXiv preprint arXiv:2205.15480*, 2022.